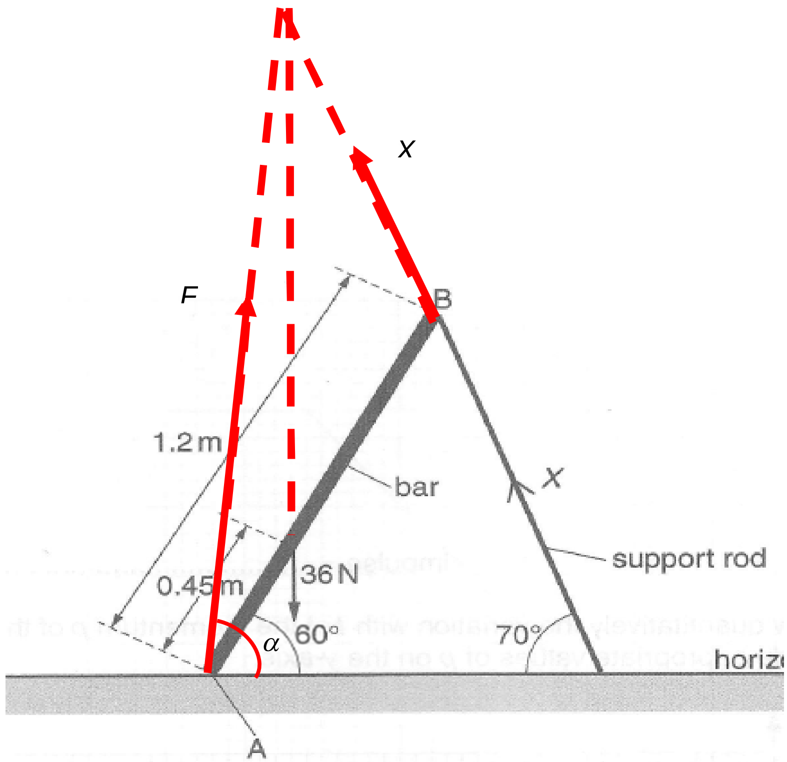
3 (a)	Taking moments about A, $X \cos 40^{\circ} (1.2) = 36 \cos 60^{\circ} (0.45)$ $X = 8.81$ N

<table>
<tr><td></td><td>


</td></tr>
<tr><td>3(b) (i)</td><td>For equilibrium, the net moments about any point must be zero. Thus, if there are no force at point A, there will be a net anti-clockwise moment about point B due to the 36 N force and then the setup will not be in equilibrium. Thus, a force must be present at A to provide a clockwise moment about B.</td></tr>
<tr><td>3(b) (ii)</td><td>For equilibrium, consider net force = 0.
Vertically:
$F \sin \alpha + X \sin 70^\circ = 36$
$F \sin \alpha = 27.7$ N ------------ (1)

Horizontally:
$F \cos \alpha = X \cos 70^\circ$
$F \cos \alpha = 3.01$ N ------------ (2)

Solving, $F = 27.9$ N</td></tr>
<tr><td>3(b)(iii)</td><td>See figure above.</td></tr>
<tr><td>4(a)</td><td>1. Most alpha-particles passed through undeflected implying that most of the gold foil is empty spaces and thus the nucleus is very small.</td></tr>
</table>